FinTrust Digital Bank — Week 2 Project Documentation

Application of the Week 1 plan · Decisions · Testing & evaluation · Limitations · Assumptions

A. Application of the Week 1 Plan (brief §3A)

Question	Response
1. What did you originally plan to accomplish?	Week 1 planned: build the analytical SQL pack; prepare and clean the data; perform exploratory analysis; build the management dashboard; identify findings; engineer predictive features; develop and evaluate a baseline risk-review model; document decisions, limitations and risks.
2. Which activities are you now implementing?	All of them, concretely: a Data Quality & Cleaning Workbook with cleaned datasets; a 10-question SQL analysis file with interpretations; a Python EDA notebook with 12 visualisations; a Power BI dashboard build; 7 documented business findings; 14 engineered features; two trained classification models with full metric evaluation.
3. Have any aspects of your approach changed?	Yes — three changes. (a) Cross-track exchange was dropped: the Week 1 brief said collaboration begins in Week 2, but the Week 2 brief explicitly states there is NO cross-track collaboration requirement and it is not assessed. (b) A second model (gradient-boosted trees) was added after the balanced logistic regression showed limited ranking headroom. (c) Accuracy was demoted as the headline metric after Week 1 showed it equals the do-nothing baseline.
4. Why did you make any changes?	(a) Primary evidence: the Week 2 brief's own policy section. (b) Evidence: both models' ROC-AUC ~0.68 and PR-AUC ~0.33 showed a ceiling worth probing with a non-linear model. (c) Evidence: Week 1's model scored 0.801 accuracy while catching only 3.9% of positive cases — accuracy was demonstrably misleading for a 19.6%-positive target.

B. Resources Used (brief §3B)

Resource	How used	Information provided	Limitation encountered
FinTrust_Customer_Data.csv / .xlsx	Loaded in Python and Excel; joined to transactions; segmented and profiled	1,500 customers, 12 fields, 0 missing, 0 duplicates	Customer_Name repeats (180 unique for 1,500) — not usable as an identifier
FinTrust_Transaction_Data.csv / .xlsx	Loaded in Python, Excel and SQL; the analysis fact table	12,000 transactions, 11 fields	192 missing cells (Device_Type 96, Location 96)
FinTrust_Data_Dictionary.xlsx	Validated column names, types and modelling suitability	12 field definitions with Modelling_Use flags	It documents Customer_Name as 'identification only' — had to be honoured
FinTrust Financial Knowledge Base	Scoped but not used for this track	GenAI track resource	Out of scope for Data Analytics Week 2

C. Tools Used (brief §3C)

Tool	Where it was used	Evidence

Microsoft Excel	Part A data-quality assessment, cleaning log, cleaned datasets, KPI sheet	FinTrust_Week2_DataQuality_Cleaning.xlsx (8 sheets)
SQL (SQLite/DuckDB syntax, portable)	Part B business analysis — 10 questions	FinTrust_Week2_SQL_Analysis.sql
Python + Pandas + NumPy	Data preparation, joins, aggregation, feature engineering	FinTrust_Week2_Data_Analysis.ipynb; week2_model.py
Matplotlib + Seaborn	12 visualisations	charts/chart1-chart12 PNG
Scikit-learn	Two classification models, metrics, cross-validation	week2_results.json
Power BI	Dashboard build (importable data + build specification supplied)	FinTrust_Week2_PowerBI_Build_Guide.docx
Git / GitHub	Version control and publication	Repository, README.md, requirements.txt

D. Testing & Evaluation Log (brief §3E and §9)

Required structure: Work Completed → Test/Evaluation → Finding → Action/Improvement → Result

Work Completed	Test / Evaluation	Finding	Action / Improvement	Result
Cleaned datasets inherited Device_Type and Location nulls	Programmatic null count on the cleaned transaction file before and after imputation	192 nulls remained in the raw file; an inner join in SQL would silently drop those 96 rows	Imputed both fields as the explicit category 'Unknown' rather than dropping records	All 12,000 records retained; location aggregates now reconcile to 12,000; imputation is auditable in the Cleaning Log
Joined Location to City for geographic analysis	Computed the agreement rate between Transaction.Location and Customer.City	They agree on only 12.7% of rows despite sharing the same 8 labels — they are different concepts	Treated them as two separate fields and documented the trap prominently	Prevented the mis-attribution of ~87% of transactions; carried into SQL Q6 as an explicit warning
Computed mean transaction value as the headline KPI	Compared mean against median and measured skewness	Skew 3.2; mean NGN 46,706 vs median NGN 10,306 — the mean overstates the typical transaction ~4.5x	Added the median to every amount KPI and added log1p(Amount) for modelling	Reported values are now defensible; models use the log-transformed amount
Baseline logistic regression (Week 1 carry-over)	Evaluated with accuracy, precision, recall, F1, confusion matrix and ROC-AUC	Accuracy 0.801 equalled the do-nothing baseline while recall was only 0.039 — accuracy was misleading	Applied class_weight='balanced', added a weighted gradient-boosted model, and tuned the decision threshold	Recall on the positive class rose from 0.039 to 0.7823 at an F1-optimal threshold of 0.4473
Model validated on a single train/test split	Five-fold stratified cross-validation on the gradient-boosted model	Folds [0.6814, 0.6801, 0.6982, 0.6458, 0.6628], mean ROC-AUC 0.6736 — consistent with the hold-out result	Reported the cross-validated mean alongside the single-split figure	Confirmed the result is not an artefact of one split; variance across folds documented
Risk-review rate compared across channels	Chi-square test of independence plus the spread of rates, not the ranking alone	Web 21.35% vs USSD 17.32% — a spread of only 4 points; the earlier test was only marginally significant (p=0.0187)	Concluded channel is a weak differentiator and declined to recommend channel-based rules on it	Findings state explicitly that absolute failure counts, not channel rates, should drive operations
Dashboard prepared for Power BI	Imported the cleaned fact and dimension tables and rebuilt the KPI cards and visuals	Power BI .pbix is a proprietary binary that cannot be authored programmatically in	Delivered the importable cleaned workbook, the .pbix-ready table structure	Dashboard is reproducible in Power BI Desktop in minutes rather than being

		this environment	and a full build specification	claimed as a native .pbix
SQL results reviewed for correctness	Cross-checked every SQL result against the independently computed Python figures	All ten results reconcile (e.g. total value NGN 560,477,354.85; 90.47% success; 19.60% review rate)	No correction required; reconciliation recorded	Two independent implementations agree, which is the strongest verification available here

E. Key Decisions (brief §3F)

Decision	Reason	Evidence	Effect on the work
Do not perform cross-track exchange	The Week 2 brief states there is no cross-track collaboration requirement and it is not assessed	Week 2 brief §1 'Important Week 2 Policy'	Removed the Week 1 assumption; the submission is entirely self-contained as required
Impute missing categorical cells rather than drop rows	Dropping 96 transactions would bias value and risk aggregates	192 missing cells, all in Device_Type and Location	12,000 records preserved; imputation logged and reversible
Keep Location and City as separate fields	They measure different things despite identical labels	12.7% agreement rate	Geographic analysis is accurate; the trap is documented for future analysts
Retain amount outliers	They are genuine positive banking values	1,474 IQR outliers (12.28%), 0 negative, 0 zero	KPIs remain unbiased; median added to protect against skew
Use Customer_ID as the only join key	Customer_Name is not unique	180 unique names for 1,500 customers; the data dictionary marks it Modelling_Use = No	No false customer collapsing
Report median alongside mean	The mean is distorted by skew	Skew 3.2; mean NGN 46,706 vs median NGN 10,306	Amount KPIs are defensible
Present all risk figures as synthetic	The brief expressly forbids presenting the label as real fraud	Week 2 brief §2 Data Disclaimer	Disclaimer carried on every artefact

F. Limitations, Assumptions & Risks (brief §3G and §10)

Type	Item	Impact	How addressed
Assumption	All FinTrust data is synthetic and educational	Findings describe a simulation, not a real bank	Disclaimer on every artefact
Assumption	Risk_Review_Flag is an unknown-provenance synthetic label	Model performance reflects the generating process, not real fraud	Never described as fraud; models labelled educational
Data quality	96 missing Device_Type and 96 missing Location	Could drop records or bias device/location comparisons	Imputed as 'Unknown'; retained and logged
Data quality	Location vs City 12.7% agreement	Geographic joins would be wrong	Fields kept separate; documented
Data quality	Amount skew 3.2	Mean-based comparisons distorted	Median reported; log transform for models
Technical	Power BI .pbix cannot be authored programmatically	A native .pbix file is not supplied	Supplied the importable cleaned data plus a full build specification and dashboard images
Technical	SQL written for SQLite/DuckDB syntax	Date functions differ on MySQL/PostgreSQL/SQL	Substitutions documented inline in the SQL file

		Server	
Methodological	Models reach ROC-AUC ~0.68 and PR-AUC ~0.33	Modest discrimination; not production-grade	Reported honestly; threshold tuned; cross-validated; Week 3 candidates identified (GBM tuning, class weights, richer features)
Risk	Class imbalance (19.6% positive)	Accuracy misleads	Balanced training, threshold tuning, PR-AUC and recall@k reported
Risk / unresolved	True fraud label has no ground truth	Predictive claims cannot be validated against reality	Framed strictly as an educational modelling exercise